

Dr.-Ing. Hujun Peng

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1991-06



EDUCATION

RWTH Aachen University	Apr 2018 - Jun 2022
Institute of Electrical Machines	Aachen, Germany
Thesis: Scalable Model-Based Energy Management Strategies for Hybrid Mobile Systems Powered by Fuel Cells and Batteries	
Technical University of Dresden	Apr 2014 - Sep 2017
Major in power electronics and electrical machinery Master	Dresden, Germany
Master's topic: Optimal design of asynchronous motors considering thermal and electromagnetic coupling	
Zhejiang University	Sep 2009 - Jun 2013
Electrical engineering and automation Bachelor	Hangzhou, China

PROFESSIONAL EXPERIENCE

RWTH Aachen University, Germany	Oct 2024 - Present
Postdoctoral researcher under Professor Simon Steentjes and Nora Leuning	
- Predicting the lifetime of insulation systems under thermal, electrical, and chemical aging. - Developing diagnostic methods for inter-turn short-circuits using transient signal analysis and machine learning. - Leading project applications and conducting research on insulation materials and systems.	
RWTH Aachen University, Germany	Jun 2022 - Jul 2024
Postdoctoral researcher under Professor Kay Hameyer	
In the Electrical Machine Materials Group, as a postdoctoral fellow in charge of the insulation laboratory, research focused on insulation system modeling and diagnostics	
RWTH Aachen University, Germany	Apr 2018 - Jun 2022
PhD researcher under Professor Kay Hameyer	
Involved in work related to energy management of fuel cell hybrid railway vehicles, high-frequency modeling of electrical machines, insulation materials testing, machine insulation system modeling, and iron materials modeling	

Teaching and Supervision Experience

- Supervised over 50 undergraduate and graduate theses - Assisted in teaching and supervising exercises for courses on Electrical Machines and Electrical Drives for seven years.	
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RESEARCH FOCUS

Machine and drive design, High-frequency modeling of electrical machines, White-box modeling of electrical machines, High-frequency modeling of power cables, Online monitoring of electrical machines and drives, Aging modeling and lifespan prediction of insulation materials.

PROJECT EXPERIENCE

Investigation of the compatibility between e-Drive coolants and electrical insulation systems under combined chemical, thermal, and electrical stresses	Feb 2026 - Jul 2026
- Project leadership and execution in collaboration with TotalEnergies. - Building on the success of previous joint projects, this follow-up project was awarded.	
Developing a fast lifetime prediction methodology for electrical machines under accelerated ageing tests	Feb 2026 - Aug 2026
This project is primarily acquired through my technical contributions and domain expertise.	

Study on the compatibility of oil-water lubricants with machine insulation system	May 2023 - Jul 2024
Project leaders and undertakers, Cooperation with TotalEnergies, Compatibility study of water-based oil lubricants with insulation materials of electrical machines	
Motor project of the eSAT hybrid air taxi	Feb 2023 - Jun 2024
Developing high-frequency models of electrical machines for monitoring motor insulation system health.	
Ferromagnetic material modeling of silicon steel sheet under the influence of machining residual stress	Oct 2023 - Apr 2024
Ferromagnetic material modeling, Project leaders and undertakers. funded by ebm-papst.	
Design of high voltage motor insulation system	Oct 2023 - Mar 2024
Develop a single-tooth high-frequency winding model to determine the maximum electric stress under inverter-fed operation	
High-frequency modeling of permanent magnet synchronous machines for electric vehicles	Feb 2021 - May 2023
Project leaders and undertakers, Cooperation with Toyota Motor Europe, High frequency modeling of PMSM, Test bench set up, Experimental measurement and data analysis	
Modeling of the properties of soft magnetic materials considering the effects of machining	Aug 2020 - Jun 2021
Modeling the influence of cutting edge effects on soft iron magnetic materials.	
Development and verification of high-power fuel cell and battery energy systems	Oct 2017 - Jun 2020
Supported by the German Ministry of Digitalization and Transport, and developed in cooperation with three research institutes within RWTH Aachen University, Ballard Power Systems and Siemens. Served as the main person in charge of the simulation design and energy management strategy development module of Siemens hydrogen fuel train drive system.	

Professional Recognition

Academic Distinction

Shortlisted for an academic position in Electric Vehicle Propulsion at Imperial College London (Feb 2025), top 4

Main Journal Papers

1. H. Peng et al., "High-Frequency Modeling of Permanent Magnet Synchronous Machines Using Grey Box Models," in IEEE Transactions on Transportation Electrification, vol. 10, no. 4, pp. 8150-8160, Dec. 2024, doi: 10.1109/TTE.2024.3363511.
2. H. Peng, Y. Yu, S. Stevic and S. Steentjes, "High-Frequency Modeling of Electrical Drive Systems Considering Test Bench Environment Influence," in IEEE Transactions on Power Electronics, vol. 40, no. 9, pp. 12816-12828, Sept. 2025, doi: 10.1109/TPEL.2025.3566700.
3. H. Peng, Y. Yu and K. Hameyer, "Conductor-Based Modeling of a Single-Tooth Winding for EMI Simulation," in IEEE Transactions on Power Electronics, vol. 39, no. 9, pp. 11099-11109, Sept. 2024, doi: 10.1109/TPEL.2024.3403669.
4. H. Peng and K. Hameyer, "A Method of Deriving an Equivalent Permeability Spectrum of Ferrous Materials Considering All Effects in Electrical Machines for 2-D Magneto-Dynamic Simulations," in IEEE Transactions on Industrial Electronics, vol. 71, no. 12, pp. 16245-16254, Dec. 2024, doi: 10.1109/TIE.2024.3390744.
5. H. Peng et al., "Voltage Distribution Modeling Along Stator Windings in Permanent Magnet Synchronous Machines Using Coil-Based Grey Box Models," in IEEE Transactions on Energy Conversion, vol. 39, no. 3, pp. 1994-2007, Sept. 2024, doi: 10.1109/TEC.2024.3370800.
6. Y. Yu, H. Peng and S. Steentjes, "A Universal Grey-Box High-Frequency Model for Calculating Bearing Voltage under WBG Inverter Drives Considering Frequency-Dependent Effects," in IEEE Transactions on Power Electronics, doi: 10.1109/TPEL.2026.3688990.
7. Peng, H., Li, J., Thul, A., Deng, K., Ünlübayir, C., Löwenstein, L. and Hameyer, K., 2020. A scalable, causal, adaptive rule-based energy management for fuel cell hybrid railway vehicles learned from results of dynamic programming. Etransportation, 4, p.100057.
8. Peng, H., Li, J., Löwenstein, L. and Hameyer, K., 2020. A scalable, causal, adaptive energy management strategy based on optimal control theory for a fuel cell hybrid railway vehicle. Applied energy, 267, p.114987.
9. Peng, H., Chen, Z., Li, J., Deng, K., Dirkes, S., Gottschalk, J., Ünlübayir, C., Thul, A., Löwenstein, L., Pischinger, S. and Hameyer, K., 2021. Offline optimal energy management strategies considering high dynamics in batteries and constraints on fuel cell system power rate: From analytical derivation to validation on test bench. Applied Energy, 282, p.116152.
10. Peng, H., Cao, H., Dirkes, S., Chen, Z., Deng, K., Gottschalk, J., Ünlübayir, C., Thul, A., Löwenstein, L., Sauer, D.U. and

- Pischinger, S., 2021. Validation of robustness and fuel efficiency of a universal model-based energy management strategy for fuel cell hybrid trains: From analytical derivation via simulation to measurement on test bench. *Energy Conversion and Management*, 229, p.113734.
11. Peng, H., Chen, Y., Chen, Z., Li, J., Deng, K., Thul, A., Löwenstein, L. and Hameyer, K., 2021. Co-optimization of total running time, timetables, driving strategies and energy management strategies for fuel cell hybrid trains. *ETransportation*, 9, p.100130.
 12. Peng, H., Chen, Z., Deng, K., Dirkes, S., Ünlübayir, C., Thul, A., Löwenstein, L., Sauer, D.U., Pischinger, S. and Hameyer, K., 2021. A comparison of various universally applicable power distribution strategies for fuel cell hybrid trains utilizing component modeling at different levels of detail: From simulation to test bench measurement. *ETransportation*, 9, p.100120.
 13. Zhang, Y., Peng, H. and Hofmann, W., 2023. Transient skin effect in highly dynamic induction motor drives: energy-optimized design. *Electrical Engineering*, 105(2), pp.1015-1024.
 14. Peng, H., Yu, Y., Stevic, S., Mönninghoff, S. and Hameyer, K., 2025. Coil-based high-frequency modeling of permanent magnet synchronous machines with single-tooth windings. *Archives of Electrical Engineering*, pp.151-163.
 15. Deng K, Peng H, Dirkes S, Gottschalk J, Ünlübayir C, Thul A, Löwenstein L, Pischinger S, Hameyer K. An adaptive PMP-based model predictive energy management strategy for fuel cell hybrid railway vehicles. *ETransportation*. 2021 Feb 1;7:100094.
 16. Deng K, Liu Y, Hai D, Peng H, Löwenstein L, Pischinger S, Hameyer K. Deep reinforcement learning based energy management strategy of fuel cell hybrid railway vehicles considering fuel cell aging. *Energy conversion and management*. 2022 Jan 1;251:115030.
 17. Deng K, Fang T, Feng H, Peng H, Löwenstein L, Hameyer K. Hierarchical eco-driving and energy management control for hydrogen powered hybrid trains. *Energy Conversion and Management*. 2022 Jul 15;264:115735.
 18. Peng, H., Li, J., Deng, K., & Hameyer, K. (2022). Machine Learning-Based Control for Fuel Cell Hybrid Buses: From Average Load Power Prediction to Energy Management. *Vehicles*, 4(4), 1365-1390.
 19. Liu, Z., Peng, H., Leuning, N., Ji, T., Zhang, P., Wu, Y., Steentjes, S., Degradation Monitoring for Motor-Drive Power Cables via Riemannian Manifold Analysis of Piecewise Impedance Spectroscopy, *IEEE Transactions on Industrial Electronics (Letters)*, under review, 2026. (Manuscript ID: 26-TIE-1057, Major Revision)

Main Conference Paper

1. Peng, H., Luo, Z., Yu, Y. and Hameyer, K., 2024, September. Modeling Voltage Distribution Along the Hairpin Windings of an Electrical Machine Using Impedance Fitting. In *2024 International Conference on Electrical Machines (ICEM)* (pp. 1-7). IEEE.
2. Hujun Peng, Niklas Driendl, Svetomir Stevic, Daniel Butterweck, Leila Sharifian, Yousef Friouz, Kay Hameyer. A Study about the Influence of Rotor Position of IPMSM on the Maximum Voltage on Motor Terminals under Inverter-fed Operation. *2023 IEEE 6th Student Conference on Electric Machines and Systems*, 2023. **(Best paper prize)**
3. H. Peng et al., "Compatibility Study of Water-Based Lubricants and Insulation Systems in Low-Voltage Electrical Machines for Automotive Applications," *2025 IEEE International Electric Machines & Drives Conference (IEMDC)*, Houston, TX, USA, 2025, pp. 123-128.
4. H. Peng, Y. Yu and S. Steentjes, "Universal Modeling Methods of Induction Machines to Study the Influence of Rotor Existence on Terminal Overvoltages During Inverter-Fed Operation," *2025 IEEE International Electric Machines & Drives Conference (IEMDC)*, Houston, TX, USA, 2025, pp. 157-163.
5. Peng, H., Li, F., Chen, Z., Deng, K., Jeschke, S. and Hameyer, K., 2022, September. Adaptive Pontryagin's Minimum Principle-Inspired Supervised-Learning-based Energy Management for Hybrid Trains Powered by Fuel Cells and Batteries. In *2022 24th European Conference on Power Electronics and Applications (EPE'22 ECCE Europe)* (pp. P-1). IEEE.
6. H. Peng, Y. Yu and S. Steentjes, "Investigation of Extracting a Transfer Function from Transient Signals During Switching Operations of Inverters for On-Line Monitoring of Electrical Machines," *2025 IEEE Workshop on Electrical Machines Design, Control and Diagnosis (WEMDCD)*, Valletta, Malta, 2025, pp. 1-6.
7. Peng, H., Li, J., Deng, K., Thul, A., Li, W., Lowenstein, L., Sauer, D.U. and Hameyer, K., 2019, October. An efficient optimum energy management strategy using parallel dynamic programming for a hybrid train powered by fuel-cells and batteries. In *2019 IEEE vehicle power and propulsion conference (VPPC)* (pp. 1-7). IEEE.

Patents

PCT/EP2024/084263: Method for diagnosing current cables, test device for diagnosing same and computer program	June 2025
DE 10 2024 105 406 A1 Method for Modeling Electrical Machines, Test Device for Modeling Same and Computer Program	August 2025

LEADERSHIP EXPERIENCE

Group Leader, Insulation Materials Laboratory

Oct 2024 - Present

- Lead research projects on the modeling of insulation system ageing behavior and lifetime prediction under combined stress factors.
- Manage daily laboratory operations, equipment maintenance, and safety compliance.
- Prepare research proposals and coordinate experimental activities.